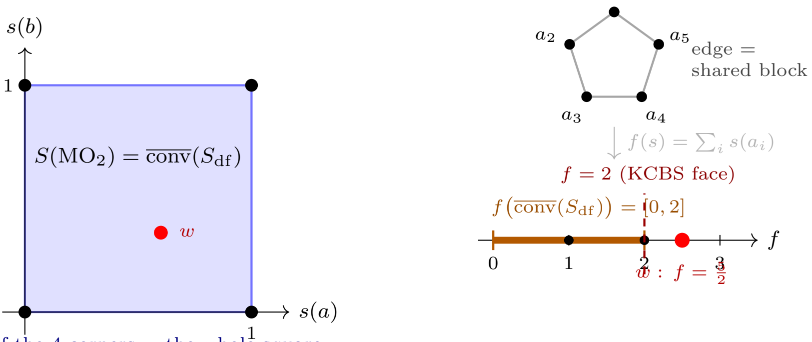
(1) MO_2 — corners fill the state space

(2) Pentagon — the lattice, and its hull projected



hull of the 4 corners = the whole square;
every state is a mixture of corners
$\Rightarrow$ **no contextual state**
(non-distributive, yet non-contextual: contextuality needs a loop, not just incompatibility — contrast panel 2)

every dispersion-free corner has $f = \sum s(a_i) \leq 2$ (odd loop: no two adjacent atoms both 1), so every mixture does; but $w \equiv \frac{1}{2}$ has $f = \frac{5}{2} \Rightarrow$ **contextual**

The spaces. The lattice L (e.g. the pentagon: 5 atoms + loop relations) is an order-theoretic object, not a vector space; it indexes the others. A *state* assigns a number to each element, so states live in $\mathbb{R}^L$ — for the pentagon, pinned by the 5 atom-values, a polytope $S(L) \subseteq [0,1]^5 \subseteq \mathbb{R}^5$. The dispersion-free states S_{df} are its $\{0,1\}$-valued *vertices*; $\overline{\mathrm{conv}}(S_{\mathrm{df}})$ is a sub-polytope, *also in* $\mathbb{R}^5$ — high-dimensional and undrawable. **The projection.** Panel (2) draws not that polytope but its image under a linear functional $g : \mathbb{R}^L \to \mathbb{R}$ (a weighted sum of atom-values; here $g = f = \sum_i s(a_i)$). Since g is linear, $g\left(\overline{\mathrm{conv}}(S_{\mathrm{df}})\right)$ is an *interval*, and $g(w)$ outside it forces w outside the hull — a one-way certificate. **Caveat.** g is a *chosen* separating functional for this w, not a canonical projection: by Hahn–Banach, w is contextual *iff some* such g sends it past the hull's image; one g detects the states violating *that* inequality, not all contextual states.